



Checkpoint I: Project Proposal

Group: G16

Date: 2025/09/15

Problem Domain

The world of Pokémon cards has evolved from a simple childhood hobby into a complex market where collectors, investors, and enthusiasts actively track trends, values, and popularity. **In the domain of Market Dynamics and Price Evolution of Pokémon Cards**, the dynamics of card prices, rarity, and demand can influence buying and selling decisions, shaping strategies for collectors and even affecting broader market behavior. Understanding these patterns offers valuable insights for anyone engaged in the hobby, from casual players deciding which packs to buy, to serious collectors evaluating long-term investments. By exploring the factors that drive card value—such as edition, popularity, and condition—our visualization aims to make these trends accessible and engaging, highlighting the interplay between nostalgia, rarity, and market dynamics in a way that is both informative and visually compelling.

Task Abstraction

1. **Are there any favorite or more popular cards per set or sets and per type or types?**
 - Abstract form: “Which items (X) are most popular regardless of category (Y)?”;
 - Variables: X = Name (nominal), Y = Card set (nominal);
 - Task type: Rank / Comparison;
 - Description: Identifies which cards are popular in each set or range of sets and in each type of range of type, regardless of release date. This helps users spot universal favorites and potential investment targets.
2. **Do rarities influence price the same way by type, by set and by year?**
 - Abstract form: “Does attribute (X) correlate with outcome (Y)?”;
 - Variables: X = Card rarity (ordinal), Y = Card price (continuous, ratio);
 - Task type: Correlation;
 - Description: Tests whether rarity level directly impacts price. This supports collectors and sellers in validating if rarity is a strong predictor of market value.
3. **Does the wear quality of a card always affect its price the same way by year and by set?**
 - Abstract form: “Does variable (X) influence variable (Y) consistently?”;
 - Variables: X = Card condition/wear quality (ordinal), Y = Card price (continuous, ratio);
 - Task type: Correlation / Distribution;
 - Examines whether the price drop between conditions follows the same pattern for all cards, or if some cards lose value much faster when condition declines. This helps reveal whether rare or iconic cards are more resilient to wear compared to common ones.
4. **Does type influence the popularity and price of a card by set and by year?**
 - Abstract form: “Does category (X) influence attribute (Y)?”;
 - Variables: X = Card type (nominal), Y = Card popularity or price (continuous ratio, ordinal);
 - Task type: Comparison / Proportion;
 - Description: Investigates whether certain Pokémon types (e.g., Fire, Water, Psychic) are consistently more popular or more valuable than others. This helps identify if market preferences align with type-based themes.

5. What are the best-selling Pokémon cards released in a year per type per rarity?

- Abstract form: “How do two related metrics (X vs Y) compare across multiple categories (Z)?”;
- Variables: X = Sales volume (quantitative, ratio), Y = Revenue (quantitative, ratio), Z = Release Date (continuous)
- Task type: Comparison / Trend;
- Description: This examines how raw sales volume compares with total revenue, broken down by year, type, rarity, and card name. It highlights whether high-volume sellers are also the biggest earners, or if certain rare/valuable cards generate more revenue despite lower sales.

Data Abstraction

Our dataset was created by merging two open sources from GitHub:

- A hierarchical dataset of Pokémon card metadata.
 - Link: [Pokémon Trading Card Game Card \(TCG\) Database for the TCGdex API](#)
 - Attributes list (21 + 9):
 - Card: **name.en**, illustrator, **rarity**, category, set, dexId, hp, **types**, evolveFrom.en, stage, abilities.type, abilities.name.en, abilities.effect.en, attacks.cost[], attacks.name.en, attacks.effect.en, attacks.damage, weaknesses.type, weaknesses.value, thirdParty.cardmarket, thirdParty.tcgplayer.
 - Set: id, **name.en**, **serie**, tcgOnline, cardCount.official, **releaseDate**, abbreviations.official, thirdParty.cardmarket, thirdParty.tcgplayer
- A price history hierarchical dataset
 - Link: [TCGdex Pricing History - Pokémon TCG Cards Database](#)
 - Attributes list (8):
 - **Condition**, **history** (dates), **avg**, **avg7**, **avg28**, **count**, **min**, **max**

The combined raw/flattened dataset contained nearly 1.68 million entries. To make the data manageable and relevant to our questions, we applied the following processing steps:

- Parsing and merging: Converted JSON/TypeScript files into tabular format (.csv/.json), then merged sources by unique card identifiers.
- Filtering by scope: Limited the dataset to the top 4 most popular series (Base, Neo, EX, Sword & Shield). This ensures focus on the most active and representative segments of the market, avoiding “long tail” sets with little trading activity.
- Attribute selection: Removed trainer, supporter, and energy cards, which follow different pricing dynamics than Pokémon cards. This guarantees consistency in the value–rarity–popularity relationship. The attributes in bold are the ones that we kept/utilized in the final dataset since they were the most relevant to answer the tasks.
- Cleaning: Dropped entries missing price or sales count information.
- Attribute refinement: Kept only the variables directly needed to answer our tasks, including release_year extracted from the original release date.

The result is a unified tabular dataset with approximately 850,000 entries and 10 attributes. Each entry corresponds to a card in a given condition, with descriptive attributes (e.g., type, rarity, set) and economic attributes (e.g., price, number of transactions(count)).

Final Attributes:

Attribute	Type	Scale	Semantics
name	Nominal	Categorical	Identifier of individual Pokémon card
set_name	Nominal	Categorical	Set name

series_name	Nominal	Categorical	Series name
types	Nominal	Categorical	Pokémon category (e.g., Fire, Water, Psychic, ...)
rarity	Ordinal	Ordered	Card rarity level (e.g., Common, Uncommon, Rare, ...)
release_year	Temporal (Continuous)	Linear, instant	Original release year of the card
price_date	Temporal (Continuous)	Linear, irregular	Date when transaction(s) were recorded; each row corresponds to one Price Date for a card
condition	Ordinal	Ordered	Wear/quality rating (e.g., Mint, NearMint, Used, ...)
avg	Quantitative (Ratio)	Sequential	Average market value of the card on a given date
count	Quantitative (Ratio)	Sequential	Number of recorded sales/transactions

Semantics and Derived Measures:

- Revenue ($\text{avg} \times \text{count}$): Computed for each row to measure total revenue per card per price date; used in tasks requiring ranking by sales or revenue.
- Price Drop Ratios by Condition: Derived measure to quantify how much value decreases when the card's condition declines (e.g., Mint $\rightarrow$ NearMint $\rightarrow$ Used); supports analysis of condition impact on price.
- release_year (derived from the original release date): Extracted to enable grouping, ranking, and trend analysis over time; used in tasks involving temporal patterns or yearly comparisons.

This dataset is sufficiently rich to answer all five tasks. It combines categorical attributes (set, type, rarity) with quantitative measures (price, sales, revenue) and temporal information (release year). The integration of descriptive and economic dimensions allows us to explore not only static comparisons but also dynamic patterns such as market evolution and value resilience over time.

Mapping

Example from our Dataset (Only represented here due to lack of space):

```
name,rarity,types,set_name,serie_name,release_year,price_date,condition,avg,count
Alakazam,Rare,Psychic,Base Set,Base,1999,2022-11-12,holo-good,2298,2
...
Hitmonchan,Common,Fighting,Neo Destiny,Neo,2002,2024-08-18,normal-good,59,16
...
```

1. Are there any favorite or more popular cards per set or sets and per type?

We want to know which cards are most popular in each set or across multiple sets, and in each type or across multiple types. So, we look at the following attributes from our dataset:

- **name** (to identify card);
- **set_name** (to group by release set);
- **types** (to group by card type);
- **count** (sales volume as measure of popularity);

And compare it with: Rankings of cards based on **count** within each **set_name** and each **type** category.

Concrete example: Group by Set + Name and by Types + Name, sum Count, then rank the cards to see which are favorites per set/type.

2. Do rarities influence price the same way by type, by set and by year?

We want to know if the effect of rarity on price is consistent across different sets, years, and types.

So, we look at the following attributes from our dataset:

- **rarity** (rarity level);
- **avg** (market value);
- **types, set_name, release_year** (for breakdowns);

And compare it with: Average prices(**avg**) by **rarity** across each **types, set_name**, and **release_year**.

Concrete example: Compute the Average price per Rarity within each Set, Types, and Release year, then test correlation between Rarity and Price separately for each grouping.

3. Does the wear (condition) quality of a card always affect its price the same way by type, by year and by set?

We want to know: Whether the effect of card condition on price is consistent across sets and years.

So, we look at the following attributes from our dataset:

- **condition** (Mint, Near Mint, Good, etc.);
- **avg** (for each condition);
- **types, set_name** and **release_year**(to check variations across **types**, sets and years);

And compare it with: Price drop ratios between **condition** levels within each **types, set_name** and **release_year**.

Concrete example: For each Set and Release Date (year), compute relative drop in Average Price from Mint → Near Mint, Near Mint → Good, etc., then compare distributions to see if the pattern holds consistently between different cards.

4. Does type influence the popularity and price of a card by set and by year?

We want to know if certain Pokémon types tend to be more popular or more valuable across sets and years. So, we look at the following attributes from our dataset:

- **types** (Fire, Water, Psychic, etc.);
- **count** (sales volume) and **avg** (value);
- **Set_name** and **release_year** (to test changes across sets/years);

And compare it with: total **count** and mean **avg** per types within each **set_name** and **release_year** to compare popularity and value

Concrete example: Group by Types + Set + Release Date (year), compute total Count and mean Average Price, then compare to see which types dominate popularity or value per set and year.

5. Pokémon Card Sales Volume vs. Revenue per Year, Type, Rarity and Name?

We want to know whether cards that sell the most (high `Count`) are also the most profitable (high revenue), across year, type, rarity, and card name. So, we look at the following attributes from our dataset:

- **count** (sales volume regardless of the condition);
- **avg** (to compute revenue regardless of the condition);
- **types, rarity, release_year, name** (for breakdowns);

And compare it with: Rankings within each breakdown (e.g., highest-volume card vs highest-revenue card in a given **release_year, type, rarity, name**).

Concrete example: For each Release Date (year), Types, and Rarity, correlate total sales (count) and revenue (count × avg) per Name, then rank to find best-sellers.